Arduino-based Vehicle Speed Detection System

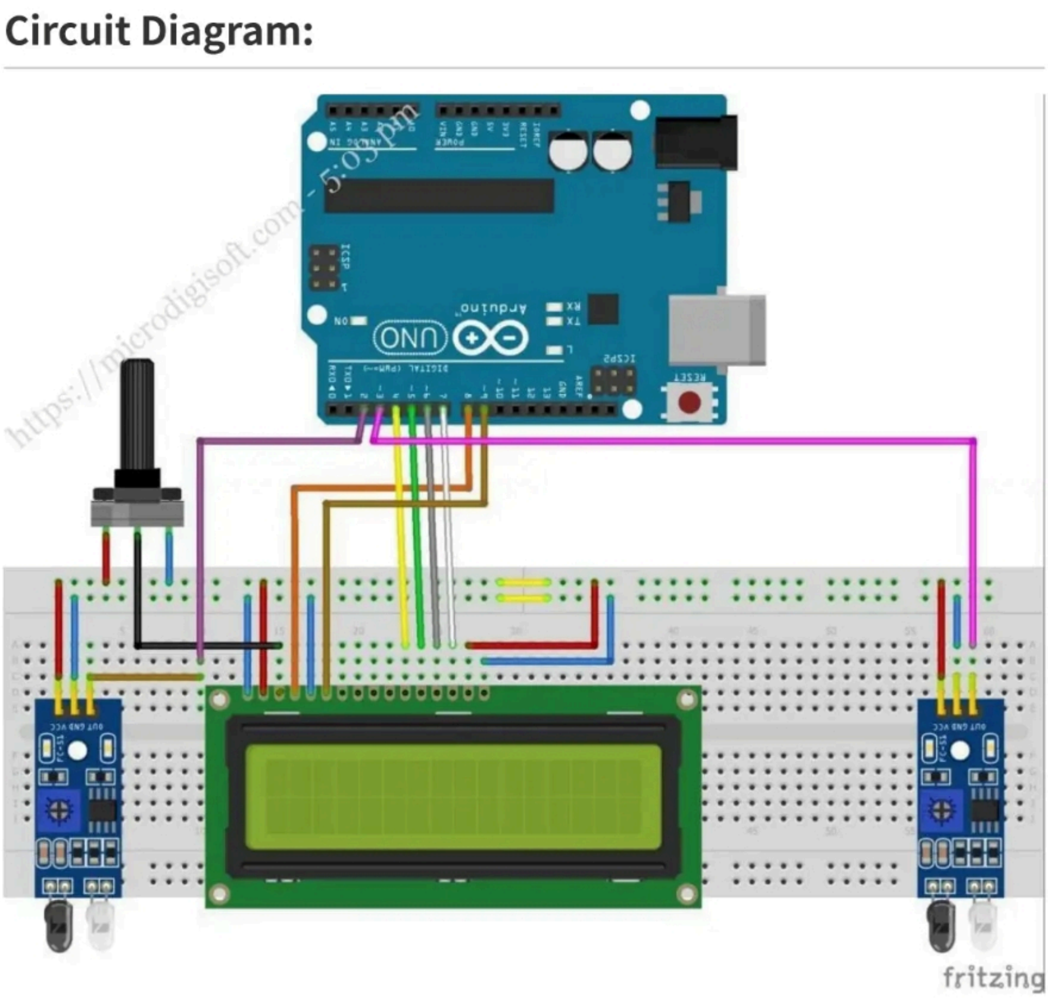
AIM:

To design and implement an Arduino-based vehicle speed detection system that accurately measures and displays the speed of a moving object (e.g., a toy car) using infrared (IR) sensors and an LCD display, providing an alert for overspeeding.

HARDWARE & SOFTWARE TOOLS REQUIRED:

S.ON	Hardware & Software requirements	QTY
1	Arduino Uno	1
2	IR (Infrared) Sensor	2
3	16x2 LCD with I2C Module	1
5	Mini Breadboard	1
6	Jumper Wires	As required
7	USB Cable (for Arduino)	1
8	Arduino IDE	Software

Circuit Diagram:



PROCEDURE:

1. Connect the components as per the circuit diagram, ensuring proper connections for the Arduino Uno, two IR sensors, the 16x2 LCD with I2C module.
2. Upload the provided code to the Arduino Uno board using the Arduino IDE.
3. Position the IR sensors at a fixed, known distance (e.g., 20 cm) apart on the simulated road. This distance is crucial for accurate speed calculation.
4. **Test the system** by passing a toy car or object between the two IR sensors. Observe the speed displayed on the LCD and verify the buzzer activates when the speed exceeds the predefined threshold (e.g., 50 Km/Hr).

5. **Calibrate the system** if necessary, by adjusting the `ACTUAL_DISTANCE` variable in the code to match the precise physical distance between the IR sensors.

PROGRAM:

```
#include <LiquidCrystal_I2C.h>
#include <Wire.h>

LiquidCrystal_I2C lcd(0x27, 16, 2);

float timer1;
float timer2;
float Time;

int flag1 = 0;
int flag2 = 0;

const float ACTUAL_DISTANCE = 0.2; // Real distance: 20 cm (0.2 m)
const float SIMULATED_DISTANCE = 5.0; // Fake distance for scaling (5 m)
float speed;

int ir_s1 = 2;
int ir_s2 = 3;
int buzzer = 11;

void setup() {
  Serial.begin(9600);
  lcd.init();
  lcd.backlight();
  pinMode(ir_s1, INPUT);
  pinMode(ir_s2, INPUT);
  pinMode(buzzer, OUTPUT);

  lcd.clear();
  lcd.begin(16, 2);
  lcd.setCursor(0, 0);
  lcd.print(" WELCOME TO ");
  lcd.setCursor(0, 1);
  lcd.print("AZIM PROJECTS   ");
  delay(2000);
  lcd.clear();
}
```

```
lcd.print("AZIM PROJECT ");
```

```
    timer1 = millis();
    flag1 = 1;
```

```
  }
  if (digitalRead(ir_s2) == LOW && flag2 == 0) {
    timer2 = millis();
    flag2 = 1;
  }

  if (flag1 == 1 && flag2 == 1) {
    if (timer1 > timer2) { Time = timer1 - timer2; }
    else if (timer2 > timer1) { Time = timer2 - timer1; }
    Time = Time / 1000; // Convert ms to seconds

    speed = (ACTUAL_DISTANCE / Time); // Speed in m/s
    speed = speed * (SIMULATED_DISTANCE / ACTUAL_DISTANCE); // Scale up
    speed = speed * 3.6; // Convert to km/h
  }

  if (speed == 0) {
    lcd.setCursor(0, 1);
    if (flag1 == 0 && flag2 == 0) { lcd.print("No Car Detected"); }
    else { lcd.print("Searching...    "); }
  }
  else {
    lcd.setCursor(0, 0);
    lcd.clear();
    lcd.print("Speed:");
    lcd.print(speed, 1);
    lcd.print("Km/Hr ");
    lcd.setCursor(0, 1);
    if (speed > 50) {
      lcd.print(" Over Speeding ");
      digitalWrite(buzzer, HIGH);
    }
    else { lcd.print(" Normal Speed  "); }
    delay(3000);
    digitalWrite(buzzer, LOW);
    speed = 0;
    flag1 = 0;
    flag2 = 0;
  }
}
```

RESULT:

The Arduino-based vehicle speed detection system successfully measures and displays the speed of a moving object.